

Association Analysis ile Kanada'daki Eyaletlerin Geleneksel - Günümüz Ekinleri ve Birliktelik Analizi

Berkay Bozlar 23290939
Furkan Günel 23290977

Proje Amacı

Bu projenin amacı, Kanada'daki eyaletlerde geçmişten günümüze tarım sepetlerinde yer alan geleneksel ve modern ekin türlerini karşılaştırmak ve bu ürünler arasındaki birliktelik ilişkilerini ortaya koymaktır.

Apriori ve FP-Growth algoritmaları kullanılarak hangi ürünlerin birlikte ekildiği analiz edilecek, böylece zamanla değişen tarım alışkanlıkları hakkında içgörüler elde edilecektir.

Veri Seti

Kullandığımız Veri Setimizin İlk Hali:

- Yıl
- Eyalet Kodu
- Ürün Çeşidi
- Ortalama Fiyatı
- Ortalama Getiri(Hektar Başına Kg)
- Toplam Üretim
- Ekili Arazi(Dönüm)
- Ekili Arazi(Hektar)
- Toplam Getiri(Dolar)

1908,AB,Barley,15,1585,84000,129800,53000,1296			
1908,AB,Flaxseed,29,950,1900,5900,2000,56			
1908,AB,Oats,18,1675,352000,519400,210000,6316			
1908,AB,"Rye, all",23,1665,5000,6500,3000,117			
1908,AB,Sugar beets,0.55,18100,38100,5200,2100,208			
1908,AB,Tame hay,0.84,5050,101000,57100,20000,846			
1908,AB,"Wheat, all",25,1690,186000,271000,110000,4617			
1908,BC,Barley,0,0,0,0,0,0			
1908,BC,Oats,0,0,0,0,0,0			
1908,BC,"Wheat, all",0,0,0,0,0,0			
1908,CA,Barley,21,1440,1016850,1745700,707100,21353			
1908,CA,"Beans, all dry (white and coloured)",59,1395,33800,59900,24200,1988			
1908,CA,Buckwheat,27,1320,155700,291300,117950,4269			
1908,CA,Corn for grain,20,3935,580600,365800,147600,11834			
1908,CA,Corn for silage,0.44,25360,2665000,259800,105100,11812			
1908,CA,Flaxseed,38,680,38100,139300,56000,1457			
1908,CA,Mixed grains,26,1650,388750,581900,235550,10140			
1908,CA,Oats,25,1205,3861650,7911100,3200550,96489			
1908,CA,"Peas, dry",31,1150,192100,412900,167050,5970			
1908,CA,"Rye, all",29,1065,43902,100300,41281,1262			

Sepet Oluşturma

```
import pandas as pd

df = pd.read_csv("../Tablolar/farm_production_dataset.csv")

df = df[['REF_DATE', 'GEO', 'Type of crop']]

df.rename(columns={'GEO': 'BÖLGE', 'REF_DATE': 'YIL'}, inplace=True)

grouped = df.groupby(['YIL', 'BÖLGE'])['Type of crop'].apply(list).reset_index()

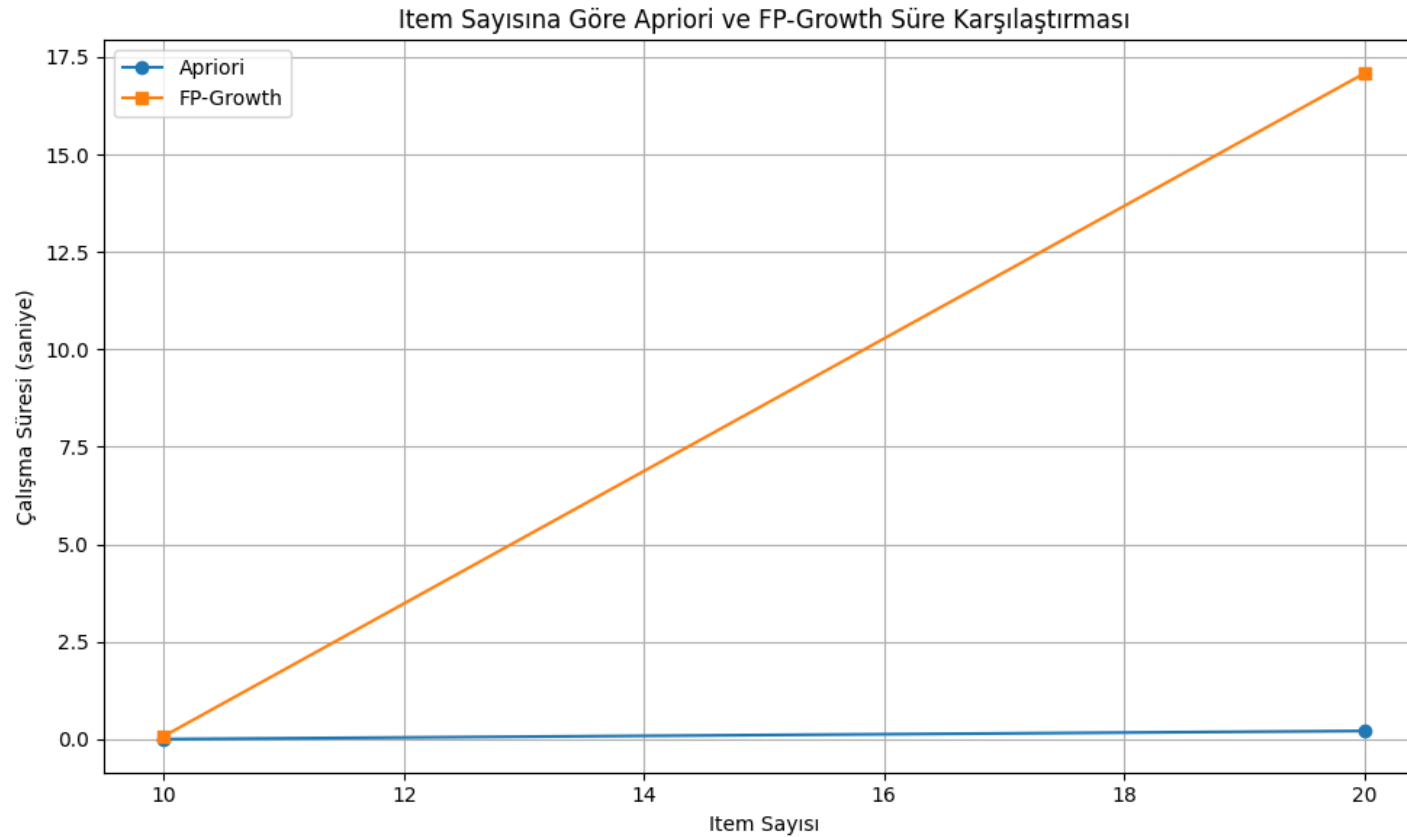
grouped.rename(columns={'Type of crop': 'SEPET'}, inplace=True)

grouped.to_csv("../Tablolar/tarim_sepetleri.csv", index=False)
```

100

[illegible]

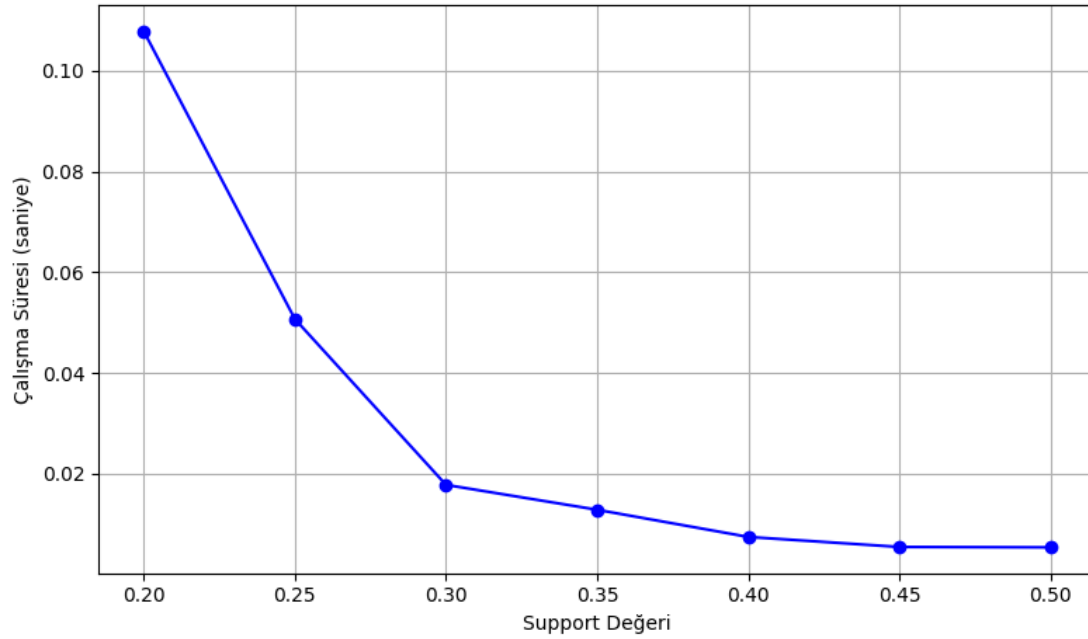
Item Sayısına Göre



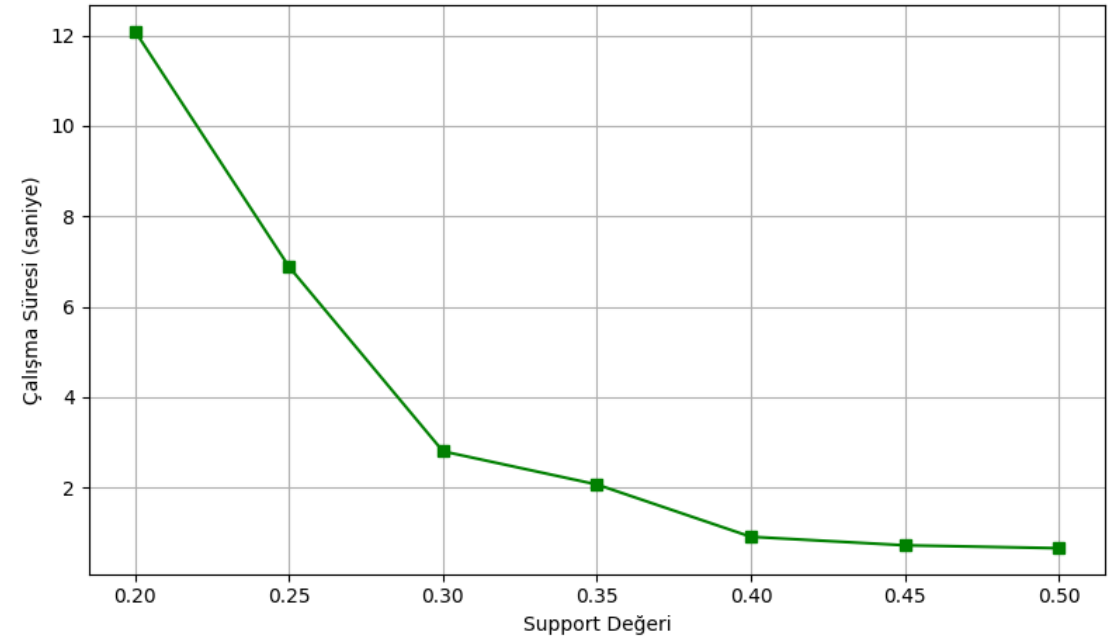
min_sup = 0.2
İçin hazırlandı

Support Değerine Göre

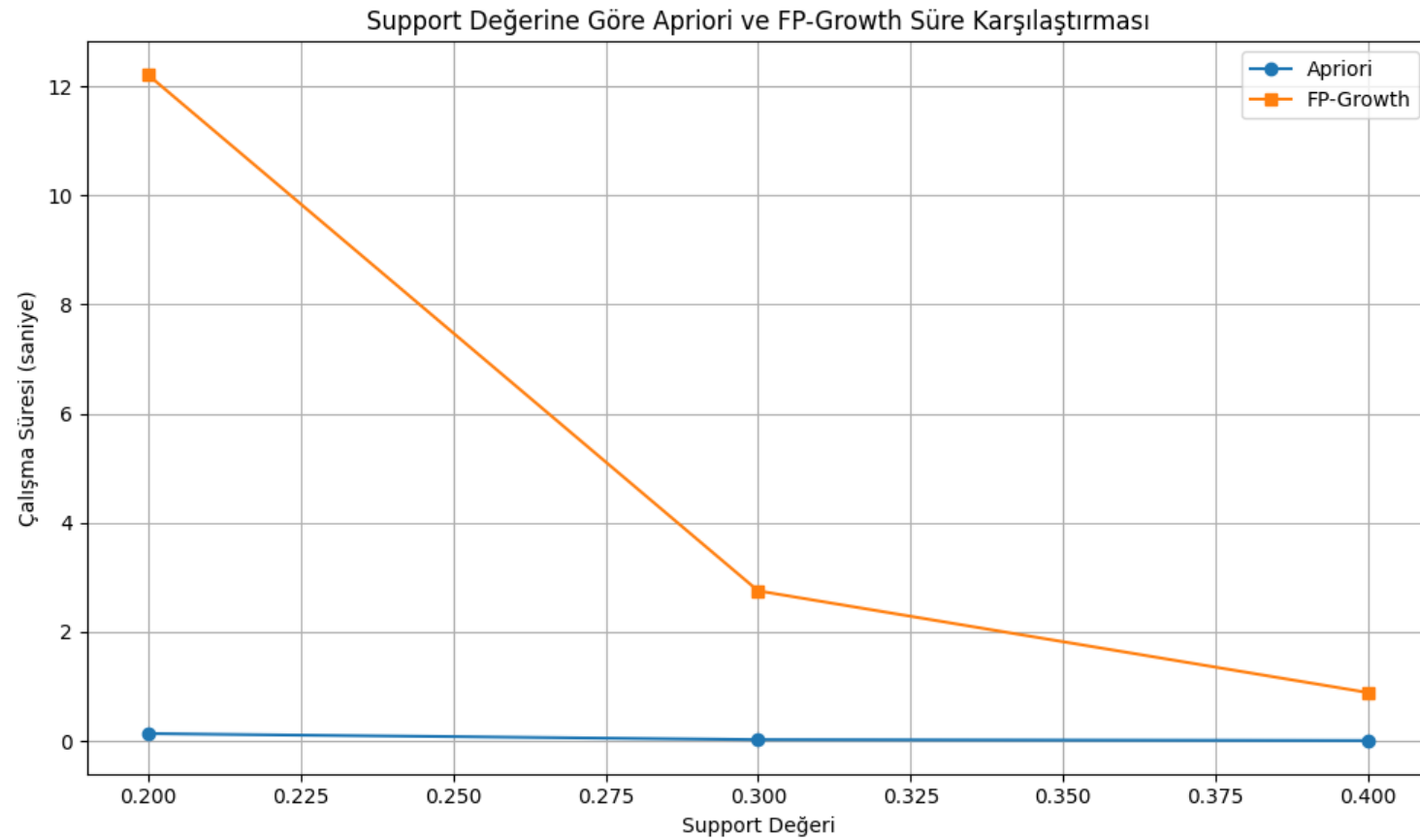
Support Değerine Göre Apriori Süresi



Support Değerine Göre FP-Growth Süresi

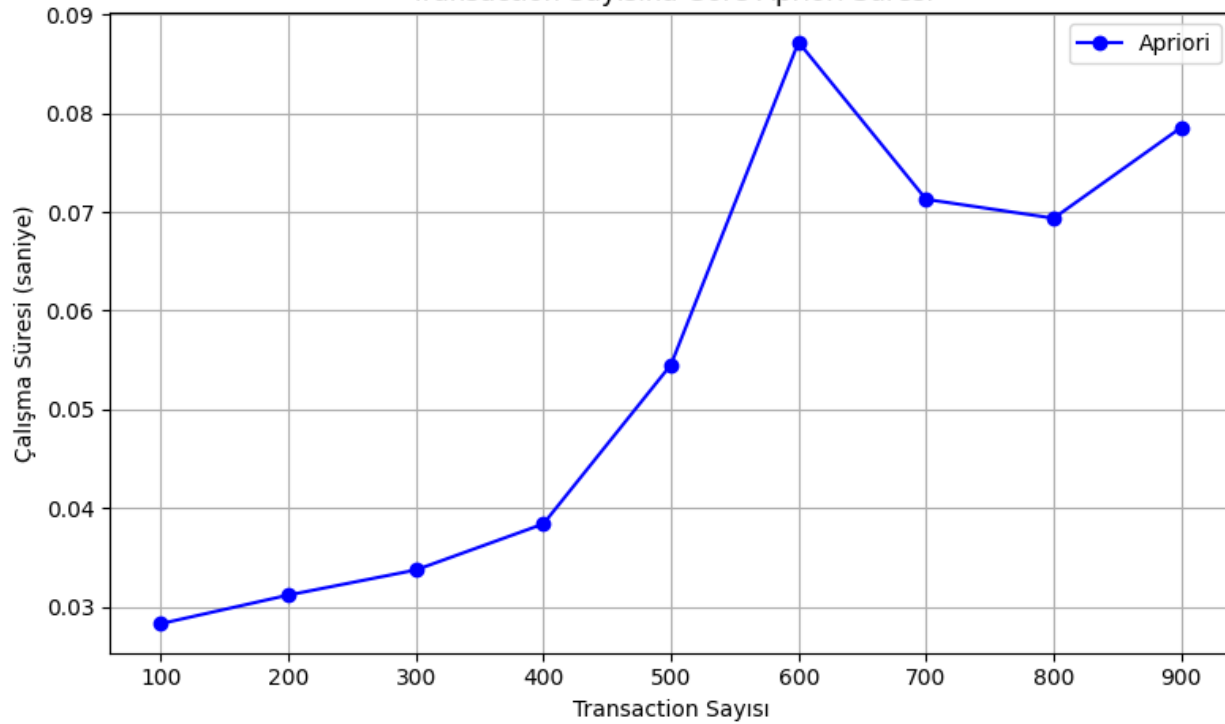


Support Deęeri Birlikte

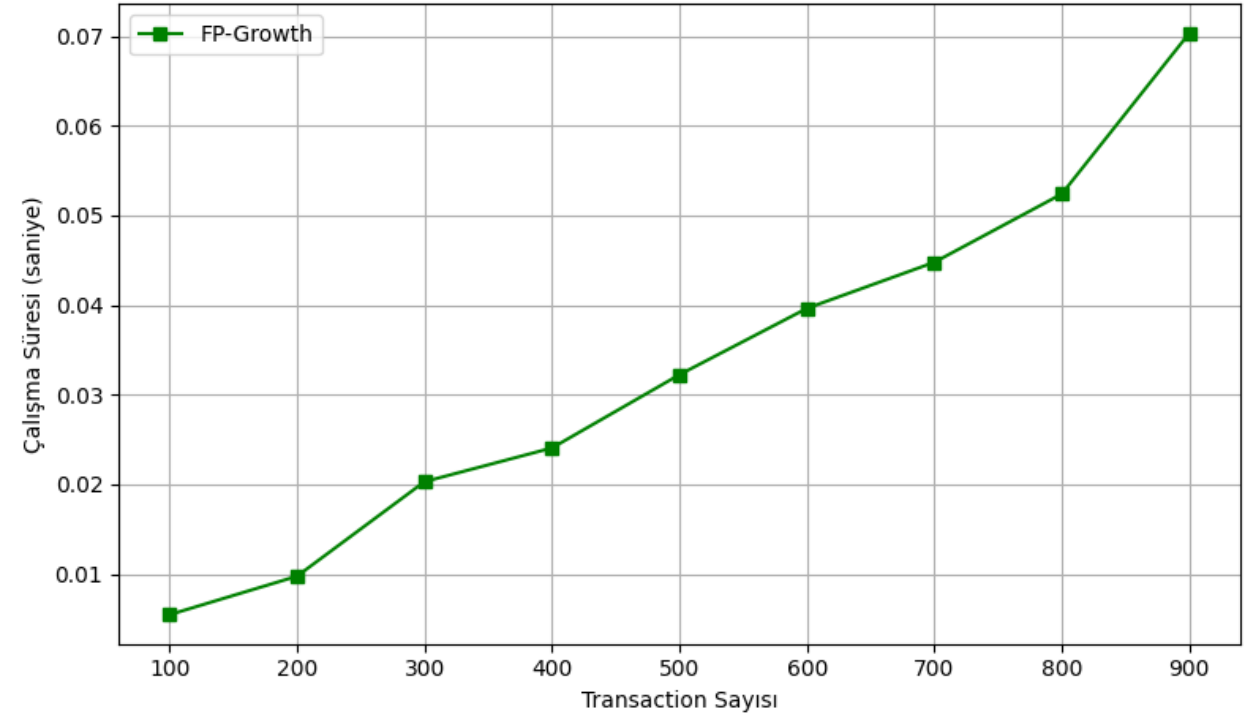


Transaction Sayısına Göre

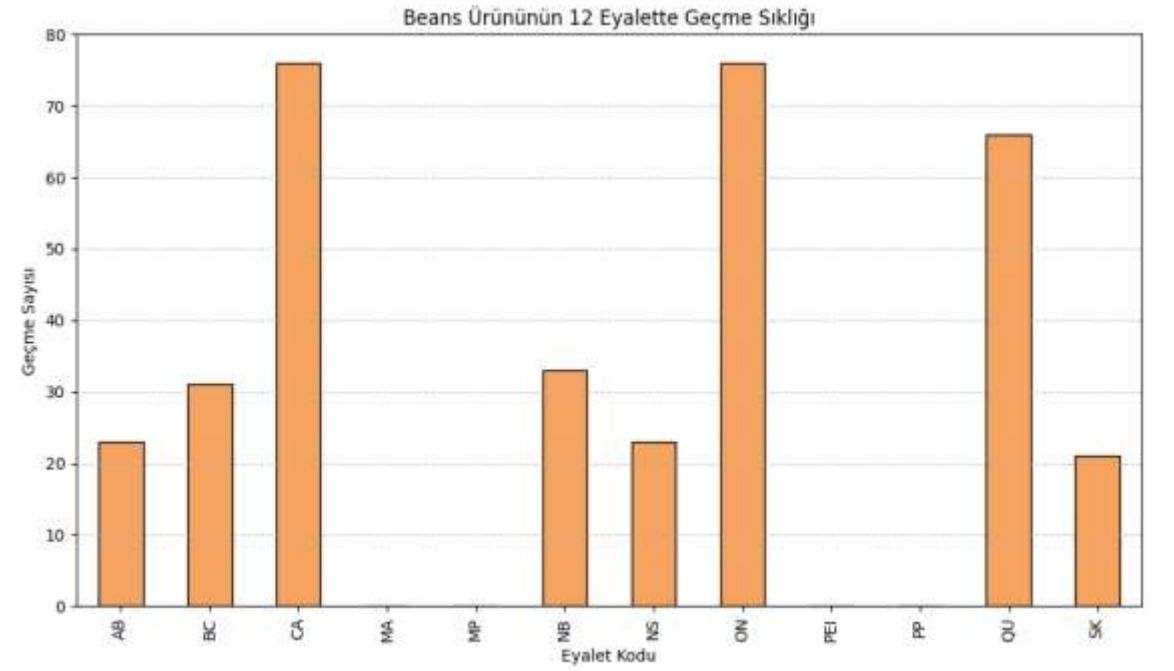
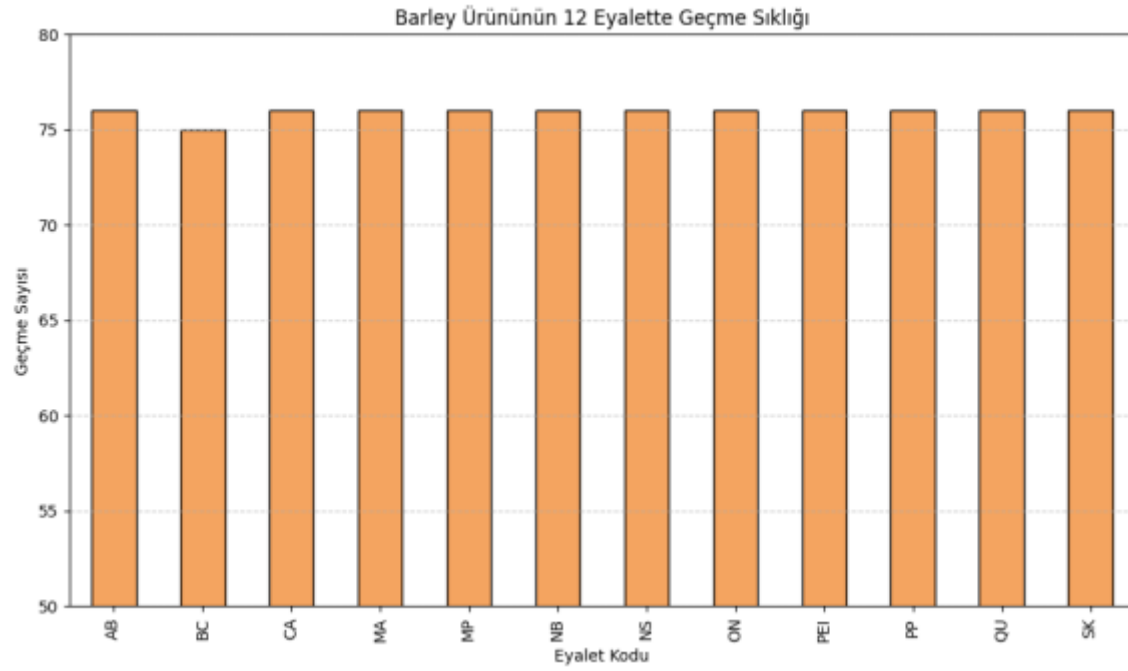
Transaction Sayısına Göre Apriori Süresi



Transaction Sayısına Göre FP-Growth Süresi



Ekim Sıklık Grafiđi



Ürünlerin Eyaletteki Ekim Sıklığı

AB Eyaleti için:

```
--- Geleneksel vs. Günümüz Ürün Sıklığı ---  
Rye, all = 67 / 7 = 9.57 | 10  
Wheat, durum = 0 / 7 = 0 | 6  
Sunflower seed = 32 / 7 = 4.57 | 0  
Beans, all dry (white and coloured) = 23 / 7 = 3.29 | 0  
Corn for silage = 65 / 7 = 9.29 | 10  
Lentils = 0 / 7 = 0 | 4  
Wheat, all = 67 / 7 = 9.57 | 10  
Wheat, spring = 0 / 7 = 0 | 6  
Mixed grains = 65 / 7 = 9.29 | 10  
Barley = 67 / 7 = 9.57 | 10  
Buckwheat = 6 / 7 = 0.86 | 2  
Oats = 67 / 7 = 9.57 | 10  
Canola (rapeseed) = 20 / 7 = 2.86 | 10  
Tame hay = 67 / 7 = 9.57 | 10  
Rye, fall remaining = 52 / 7 = 7.43 | 10  
Mustard seed = 25 / 7 = 3.57 | 10  
Corn for grain = 0 / 7 = 0 | 6  
Peas, dry = 65 / 7 = 9.29 | 10  
Wheat, winter remaining = 0 / 7 = 0 | 7  
Sugar beets = 67 / 7 = 9.57 | 10  
Flaxseed = 67 / 7 = 9.57 | 10  
Rye, spring = 52 / 7 = 7.43 | 10
```

İlk 70 Yıl için Birliktelik Analizi

Apriori Çıktısı

```
antecedent,consequent,support,confidence,lift
Sugar beets,Flaxseed,0.4,0.99,1.55
Flaxseed,Sugar beets,0.4,0.63,1.55
"Peas, dry",Sugar beets,0.4,0.54,1.34
Sugar beets,"Peas, dry",0.4,0.99,1.34
Flaxseed,"Peas, dry",0.63,0.99,1.33
"Peas, dry",Flaxseed,0.63,0.85,1.33
"Beans, all dry (white and coloured)","Peas, dry",0.4,0.97,1.3
"Peas, dry","Beans, all dry (white and coloured)",0.4,0.54,1.3
"Rye, all",Flaxseed,0.64,0.83,1.29
Flaxseed,"Rye, all",0.64,1.0,1.29
```

Fp-Growth Çıktısı

```
antecedent,consequent,support,confidence,lift
Flaxseed,Sugar beets,0.4,0.63,1.55
Sugar beets,Flaxseed,0.4,0.99,1.55
Sugar beets,"Peas, dry",0.4,0.99,1.34
"Peas, dry",Sugar beets,0.4,0.54,1.34
Flaxseed,"Peas, dry",0.63,0.99,1.33
"Peas, dry",Flaxseed,0.63,0.85,1.33
"Peas, dry","Beans, all dry (white and coloured)",0.4,0.54,1.3
"Beans, all dry (white and coloured)","Peas, dry",0.4,0.97,1.3
Sugar beets,"Rye, all",0.4,1.0,1.29
"Rye, all",Sugar beets,0.4,0.52,1.29
```

min_sup = 0.4 için

Apriori süresi: 0.1081 saniye

FP-Growth süresi: 1.1635 saniye

Son 10 Yıl için Birliktelik Analizi

Apriori Çıktısı

```
antecedent,consequent,support,confidence,lift
Flaxseed,"Peas, dry",0.46,1.0,2.18
"Peas, dry",Flaxseed,0.46,1.0,2.18
Flaxseed,Canola (rapeseed),0.43,0.93,1.83
Canola (rapeseed),Flaxseed,0.43,0.84,1.83
Canola (rapeseed),"Peas, dry",0.43,0.84,1.83
"Peas, dry",Canola (rapeseed),0.43,0.93,1.83
"Rye, fall remaining",Canola (rapeseed),0.49,0.71,1.41
Canola (rapeseed),"Rye, fall remaining",0.49,0.97,1.41
Corn for grain,"Rye, fall remaining",0.47,0.97,1.4
"Rye, fall remaining",Corn for grain,0.47,0.68,1.4
```

Fp-Growth Çıktısı

```
antecedent,consequent,support,confidence,lift
"Peas, dry",Flaxseed,0.46,1.0,2.18
Flaxseed,"Peas, dry",0.46,1.0,2.18
"Peas, dry",Canola (rapeseed),0.43,0.93,1.83
Canola (rapeseed),"Peas, dry",0.43,0.84,1.83
Canola (rapeseed),Flaxseed,0.43,0.84,1.83
Flaxseed,Canola (rapeseed),0.43,0.93,1.83
"Rye, fall remaining",Canola (rapeseed),0.49,0.71,1.41
Canola (rapeseed),"Rye, fall remaining",0.49,0.97,1.41
"Rye, fall remaining",Corn for grain,0.47,0.68,1.4
Corn for grain,"Rye, fall remaining",0.47,0.97,1.4
```

min_sup = 0.4 için

Apriori süresi: 0.0246 saniye

FP-Growth süresi: 0.6232 saniye

Çıkarım

İlk 70 yılda en sık birlikte görülen ürün çiftleri arasında "Sugar beets", "Flaxseed" ve "Peas, dry" öne çıkarken; son 10 yılda bu yapı değişmiş, "Flaxseed", "Peas, dry" ve "Canola (rapeseed)" arasında çok güçlü birliktelikler gözlemlenmiştir.

Apriori ve FP-Growth algoritmaları neredeyse aynı sonuçlar üretmiş, bu da bulguların güvenilirliğini arttırmıştır.

Tüm Yılların Birliktelik Analizi (AB)

Apriori Çıktısı

```
antecedent,consequent,support,confidence,lift
Rye, spring,Sunflower seed,0.41,0.51,1.25
Rye, fall remaining,Sunflower seed,0.41,0.51,1.25
"Mustard seed","Rye, fall remaining",0.45,1.0,1.25"
Rye, spring,Mustard seed,0.45,0.56,1.25
"Mustard seed","Rye, spring",0.45,1.0,1.25"
Rye, fall remaining,Mustard seed,0.45,0.56,1.25
"Rye, fall remaining","Rye, spring",0.8,1.0,1.25"
"Sunflower seed","Rye, fall remaining",0.41,1.0,1.25"
"Rye, spring","Rye, fall remaining",0.8,1.0,1.25"
"Sunflower seed","Rye, spring",0.41,1.0,1.25"
```

Fp-Growth Çıktısı

```
antecedent,consequent,support,confidence,lift
Rye, fall remaining,Sunflower seed,0.41,0.51,1.25
Rye, spring,Sunflower seed,0.41,0.51,1.25
"Sunflower seed","Rye, fall remaining",0.41,1.0,1.25"
"Sunflower seed","Rye, spring",0.41,1.0,1.25"
"Mustard seed","Rye, spring",0.45,1.0,1.25"
Rye, spring,Mustard seed,0.45,0.56,1.25
"Rye, fall remaining","Rye, spring",0.8,1.0,1.25"
"Rye, spring","Rye, fall remaining",0.8,1.0,1.25"
"Mustard seed","Rye, fall remaining",0.45,1.0,1.25"
Rye, fall remaining,Mustard seed,0.45,0.56,1.25
```

$\text{min_sup} = 0.4$ için

Apriori süresi: 10.2439 saniye

FP-Growth süresi: 12.1966 saniye

Tüm Yılların Birliktelik Analizi (AB)

Apriori Çıktısı

```
antecedent,consequent,support,confidence,lift
Canola (rapeseed),Mustard seed,0.39,1.0,2.24
Mustard seed,Canola (rapeseed),0.39,0.88,2.24
Sunflower seed,Mustard seed,0.32,0.77,1.73
Mustard seed,Sunflower seed,0.32,0.71,1.73
Rye, spring,Sunflower seed,0.41,0.51,1.25
Rye, fall remaining,Sunflower seed,0.41,0.51,1.25
"Canola (rapeseed),"Rye, fall remaining",0.39,1.0,1.25"
"Canola (rapeseed),"Rye, spring",0.39,1.0,1.25"
"Mustard seed,"Rye, spring",0.45,1.0,1.25"
Rye, fall remaining,Mustard seed,0.45,0.56,1.25
```

Fp-Growth Çıktısı

```
antecedent,consequent,support,confidence,lift
Canola (rapeseed),Mustard seed,0.39,1.0,2.24
Mustard seed,Canola (rapeseed),0.39,0.88,2.24
Mustard seed,Sunflower seed,0.32,0.71,1.73
Sunflower seed,Mustard seed,0.32,0.77,1.73
Rye, spring,Sunflower seed,0.41,0.51,1.25
Rye, fall remaining,Sunflower seed,0.41,0.51,1.25
"Rye, fall remaining,"Rye, spring",0.8,1.0,1.25"
"Rye, spring,"Rye, fall remaining",0.8,1.0,1.25"
"Sunflower seed,"Rye, spring",0.41,1.0,1.25"
"Sunflower seed,"Rye, fall remaining",0.41,1.0,1.25"
```

min_sup = 0.3 için

Apriori süresi: 28.0626 saniye

FP-Growth süresi: 40.3478 saniye

Çıkarım

FP-Growth'un daha küçük support değerlerinde daha fazla ilişki bulma eğiliminde olduğunu gösteriyor. Apriori ise düşük support değerlerinde bazı kombinasyonları atladığı için farklı sonuçlara ulaşabiliyor.

Sonuç olarak, düşük min_support değerleri kullanıldığında FP-Growth daha geniş kapsamlı ilişkileri ortaya çıkarabilir.

Özetle

Kriter

Çalışma Mantığı

Hız & Verimlilik

Küçük Min_Support Etkisi

Bellek Kullanımı

Kuralların Güvenilirliği

Apriori

İteratif olarak sık öğeleri belirler

Büyük veri setlerinde daha yavaş

Daha az ilişki çıkarabilir, bazılarını atlar

Daha fazla bellek tüketir

Daha dengeli ve az sayıda ilişki verir

FP-Growth

FP-Tree yapısı kullanarak veri tarar

Büyük veri setlerinde daha hızlı

Daha fazla ilişki bulur, kapsam genişler

Daha az bellek kullanır

Daha fazla ilişki, ancak bazıları zayıf olabilir



DİNLEDİĞİNİZ İÇİN TEŞEKKÜRLER